

CUSTOMER MILESTONE BRIEF

MS #40 - IMS Qualification Hardening

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.40.0
Date	2026-08-31
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #40 - IMS Qualification Hardening converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Qualifies bounded IMS database structure, navigation, status, mutation, checkpoint, and restart semantics.

Why this matters to a customer

Qualifies bounded IMS database structure, navigation, status, mutation, checkpoint, and restart semantics.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Bound IMS qualification to the canonical graph inventory of four databases, four dataset groups, four PSBs, six PCBs, three segments, three fields, and their structural relationships.
- Added a deterministic 40-case synthetic semantic corpus with 32 targeted boundary cases, 36 passing cases, four explicit fail-closed cases, and 94 observed feature categories.
- Covered HIDAM, secondary-index and GSAM boundaries; GU/GN/GNP and GHU/GHN/GHNP navigation; qualified, unqualified, multi-level, and invalid SSAs; blank, GE, GB, II, AJ, and DJ status codes; ISRT, REPL, DLET, PROCOPT, SENSEG, CHKP, XRST, ROLB, and scheduling boundaries.
- Added a 28-entry IMS compatibility ledger using all five governing classifications and explicit exclusions for Fast Path DEDB, MSDB, IMS TM shared queues, DBRC logging, and recovery.

- Separated eleven gates for graph inventory, corpus provenance, DBD/storage, hierarchy and SSAs, PSB/PCB policy, navigation/status, mutation behavior, checkpoint/recovery, runtime exclusions, the existing private expiry-purge proof, and authorized native execution.
- Added frozen schemas, POSIX/PowerShell release-gate integration, deterministic receipts, graph holdouts, exact-corpus checks, negative diagnostics, and rehashed-overclaim tests.
- Kept native IMS, IMS TM, Fast Path, DBRC recovery, restart, runtime, mainframe-equivalence, and production claims false pending authorized region, database, logging, and recovery evidence.

How it differs from earlier work

MS #40 builds on MS #39 (CICS and VSAM Qualification Hardening). The earlier milestone established its own bounded capability; this milestone adds ims qualification hardening while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use ims qualification hardening as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Kept native IMS, IMS TM, Fast Path, DBRC recovery, restart, runtime, mainframe-equivalence, and production claims false pending authorized region, database, logging, and recovery evidence.

Source of record

- CHANGELOG.md release section(s): 0.40.0
- README.md product and operating guidance

- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.